Important Questions

1. Explain the concept of Machine Learning with real-life applications.
2. Why is Machine Learning important in today's technological world? Discuss with examples.
3. Compare and contrast Supervised, Unsupervised, and Reinforcement Learning.
4. What are the major uses of Machine Learning in different industries? Provide examples.
5. Explain the importance of Python in Machine Learning. List and describe essential libraries used in ML.
6. How do you install and use libraries like NumPy, Pandas, and Matplotlib in Python? Provide relevant code snippets.
7. Explain the role of NumPy and SciPy in data manipulation and computation. Provide examples.
8. Differentiate between qualitative and quantitative data with examples.
9. Explain the difference between nominal, ordinal, discrete, and continuous data types.
10. What are the different data types in Pandas? Explain with examples.
11. Describe data splitting and data scaling. Why are they important in Machine Learning?
12. How do you handle missing values in a dataset? Explain with Python code.
13. Explain the concepts of mean, median, mode, standard deviation, and variance with examples.
14. What is the goal of Exploratory Data Analysis (EDA)? Explain its importance in Machine Learning.
15. Explain the different types of Exploratory Data Analysis techniques. Provide examples.
16. What is time series analysis? How is it useful in Machine Learning? Explain with an example.
17. How do you get insights about a dataset using Python? Explain with functions from Pandas.
18. Explain the role of data visualization in EDA. List different types of plots used in data visualization.
19. Describe how missing data is handled in Exploratory Data Analysis. Provide relevant Python functions.
20. What are some common data visualization tools used in EDA? Explain with Python examples.
21. What is supervised learning? Explain classification and regression with examples.
22. Explain Decision Trees in detail. How do they work in supervised learning?
23. What is clustering? Explain its advantages and disadvantages with an example.
24. What is Reinforcement Learning? Explain its applications and how it differs from Supervised and Unsupervised Learning.
25. What is Semi-Supervised Learning? How is it different from Supervised and Unsupervised Learning?
26. How does clustering work? Explain different clustering techniques used in Unsupervised Learning.